

# Introduction

In 2016, one of the most famous and most exemplary DDoS attacks happened; more than a million devices attacked DNS provider Dyn, resulting in major online services like Netflix, Airbnb, and the BBC being cut off. More recently, social media service Bluesky was attacked for 24 hours on April 15th by a pro-Iran hacktivist group. Distributed Denial of Service attacks, or DDoS attacks are attempts to make digital infrastructure unavailable by flooding them with requests from multiple sources.

Denial of Service attacks are (one of the) most common network security threat experienced by Enterprises, Governments and Educational bodies, while the customers of Service Providers (e.g. ISPs and Data Centers) remain to be the number one target of DDoS attacks [1] while network-layer attacks target vital Internet structure [2]. DDoS attacks are easy to execute, even without technical knowledge [3]. AI integrated attacks and increasing sophistication only prove that DDOS attacks are becoming more dangerous: A record of 31.4 terabytes per second attack has been stopped by an Internet-service provider [4].

Due to the relevance of DDoS attacks, many researches have been conducted on detection, defending, and mitigating attacks. In this research, we investigate the severity of a DDoS attack on its societal impact, where we use Google Alerts as a source to identify DDoS attacks. These alerts are collected in a database, which has been annotated as “*Arrest*”, “*Attack*”, “*Research*”, and “*Other*” in prior research [5].

We want to see how several dimensions influence the social ‘popularity’ of a DDoS attack. This creates an insight into public perception of DDoS related events, and how this changes historically, as well as gives insight into socioeconomic damage perceived through these events. Possibly, this insight can be used to make more educated decisions in a cost/benefit analysis on cybersecurity.

This research records socioeconomic dimensions in DDoS related events (news articles, blogs, etc). The dimensions we record, are: Political, Economical, Legal, Social, and Technical impact. A common analysis models which combines these factors, is the PESTLE analysis model, which will be used in this research. Based on this analysis model, the research aims to create a tool to visualize the insight gained from this dataset.

The main goals of this research are:

- Analyse annotated Google alert dataset with PESTLE analysis model
- Create valuation model based on analysis
- Expand current annotated Google alert dataset with analysis data

- A tool that can generate insight into the (analysed) impact of a DDoS attack given a dataset

This research is comprised of two main research questions:

RQ1: What is the impact of PESTLE dimensions on the news popularity of a DDOS attack?

RQ2: How do we create a measurement tool to contextualize/visualize the impact (based on the annotated dataset)?

## Methodology and approach

The dataset used in this research has already been collected. This dataset, comprised of Google Alerts, has been described and collected by Abhishta [6]. The dataset consists of Google Alerts on the keywords “DDoS” and “Distributed Denial of Service”. The dataset has been annotated as “*Arrest*”, “*Attack*”, “*Research*”, and “*Other*” in research by Wim Kamerman.

This research uses analysis on article level (per datapoint) and on event level (per DDoS event). This research performs a manual, qualitative analysis on PESTLE dimensions occurring in the dataset on article level. We extract the relevant features in Google Alert notifications and store this metric in the dataset. Each PESTLE dimension is operationalized using observable indicators extracted from the text. Examples of this, could be; targeting of government (Political), public service provider targeting (Social), attack size mentioned (Technological), or the damage done is mentioned (Economical).

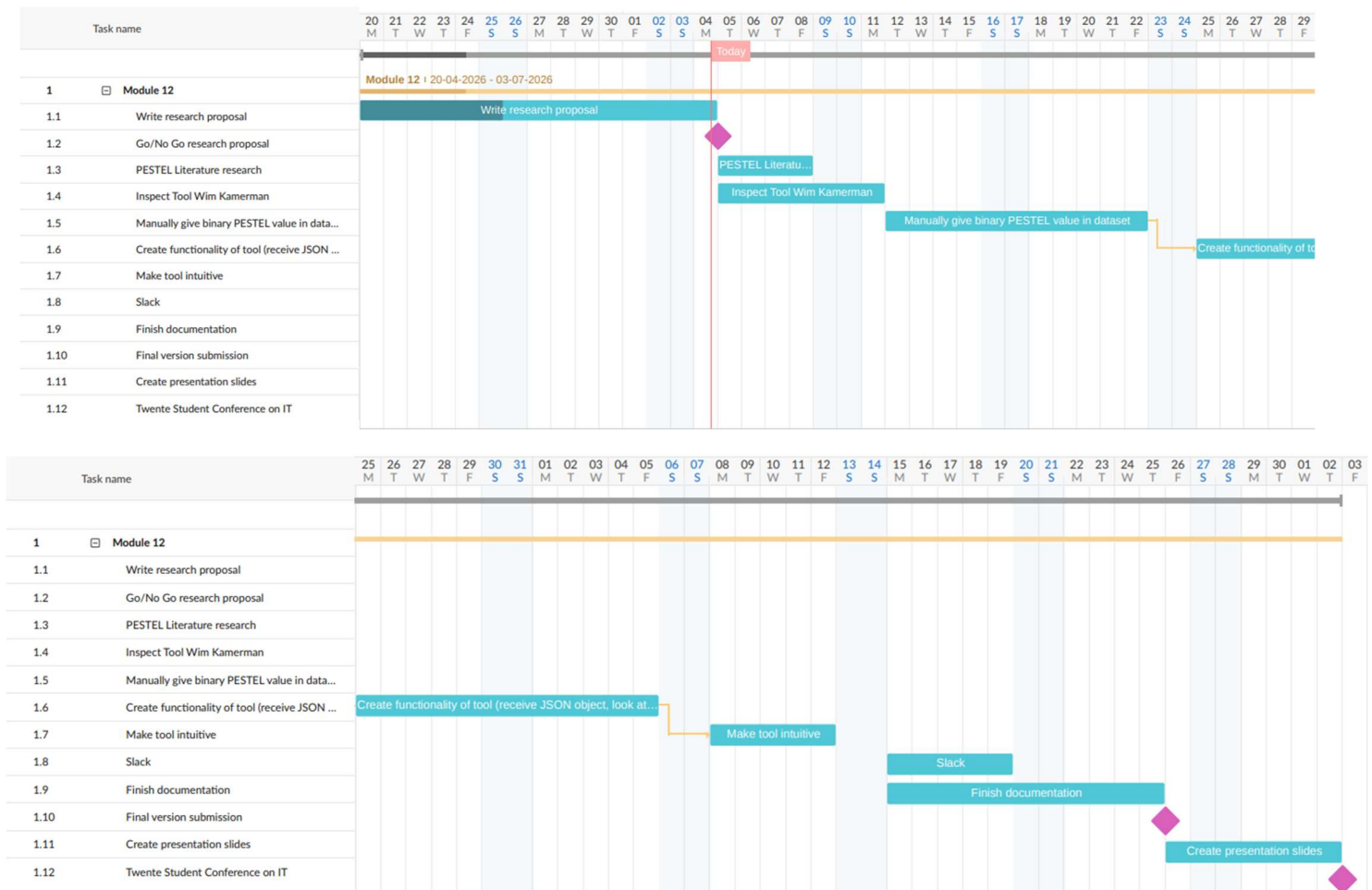
We operationalize the ‘popularity’ of an event as the number of articles referring to the same DDoS event within a defined timeframe. A subset of the dataset will be manually annotated and used to train a supervised text classification model to automatically detect PESTLE-related features. This process can be assisted via a local Large Language Model provided by the University of Twente to help code this artifact, and by using the previous research from Kamerman. The Machine Learning and text classification models for text annotation, provided by Kamerman, will be used and expanded. A regression analysis will be conducted to determine the relationship between PESTLE dimensions and the popularity of DDoS events.

## Work programme

This module consists of 11 weeks. In this research, there are two substantial content-related tasks which need to be completed in the given timeframe:

- Give PESTEL value to dataset
- Create tool to gain insight into ‘value’ of alerts about DDOS

Next to that, we also need time for documentation (in-code and in the paper), preparing the presentation for the *Twente Student Conference on IT*, and we need time for setbacks. Inspired by the Student Manual, which states a 2 week proposal stage and a two week conference prepare stage, we gain the following schedule:



During the paper writing part, we divided the time as follows. The first week, time has to be made clear for setup and literature research. Since the research this paper is based on is from 2018, the libraries and code written are outdated, which will likely give some problems early on. Manually going through (a sample of) the dataset will be labour intensive. This process could be sped up using large language models, but a human viewpoint is probably preferred, which is why we think 2 weeks will be necessary to properly perform this task. Creating the tool, and making it intuitive and usable, will likely cost three weeks of time. A large part of this process can only be started once the dataset has been annotated, which is why it is essential we do the annotation correctly.

In the end, we have reserved 2 weeks of time for documenting purposes, as well as slack and review moments. It is likely that review moments are spread throughout the module; for simplification reasons we have grouped them together in the last weeks. In these last weeks, the conference needs to be prepared with slides, and the presentation must be practiced thoroughly. This will likely take another week. Together this research spans the entire module.

## Bibliography

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